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MOBILE APPLICATION PROGRAMMING

GROUP NAME:

SCRUMSHANK REDEMPTION

Final Project Report:

PetPal – Smart Pet Care Management Mobile Application

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Executive Summary

PetPal is a comprehensive smart pet care management application that connects pet owners, veterinarians, and pet sitters in a unified ecosystem. The application leverages modern mobile technology (Flutter), cloud infrastructure (Firebase), and intelligent design patterns to provide a seamless experience for managing all aspects of pet care.

1.0 Stakeholder and User Identification

1.1 Primary Stakeholders

Pet Owners

Description: Individual users who own one or more pets and need to manage their pets' health, activities, and care services

Goals:

- Keep comprehensive health and activity records for their pets
- Schedule and manage veterinary appointments
- Find and book reliable pet sitters
- Track pet hotel reservations
- Monitor pet wellness through activity logs and health reports

Pain Points: Scattered information across multiple platforms, difficulty tracking medical history, finding trustworthy care providers

Veterinarians

Description: Licensed veterinary professionals offering medical services and consultations

Goals:

- Manage patient appointments efficiently
- Access pet medical history and health records
- Communicate with pet owners regarding health concerns
- Maintain professional records and schedules

Pain Points: Manual appointment scheduling, lack of unified patient records, communication delays with pet owners

Pet Sitters & Care Providers

Description: Professional pet sitters and care service providers offering daily care, grooming, and boarding services

Goals:

- Build reputation and client base
- Manage bookings and schedules efficiently
- Communicate with pet owners about care details
- Showcase services and availability

Pain Points: Reliance on informal platforms, scattered client information, difficulty managing multiple bookings

Pet Hotel/Boarding Facility Operators

Description: Professional facilities offering boarding and hotel services for pets

Goals:

- Manage pet check-ins/check-outs efficiently
- Track pet care activities and health during stays
- Communicate with owners about pet welfare
- Maintain records of guest pets

Pain Points: Manual check-in processes, difficulty tracking multiple pets, communication gaps with owners

1.2 Secondary Stakeholders

- Firebase Cloud Services: Infrastructure and data backend
- Mobile Device Users: iOS, Android, and Web users
- App Store/Play Store: Distribution channels
- Insurance Companies: May use health records for pet insurance claims

1.3 User Personas

Persona 1: Sarah Chen - Busy Professional Pet Owner

- Age: 32, works in tech
- Has 2 dogs and 1 cat
- Needs to schedule vet appointments, track vaccination records
- Frequently travels and needs reliable pet sitter services
- Tech-savvy and expects a modern, intuitive app

Persona 2: Dr. James Wilson - Veterinarian

- Age: 45, operates small animal clinic
- Manages 15-20 appointments per day
- Needs quick access to patient history

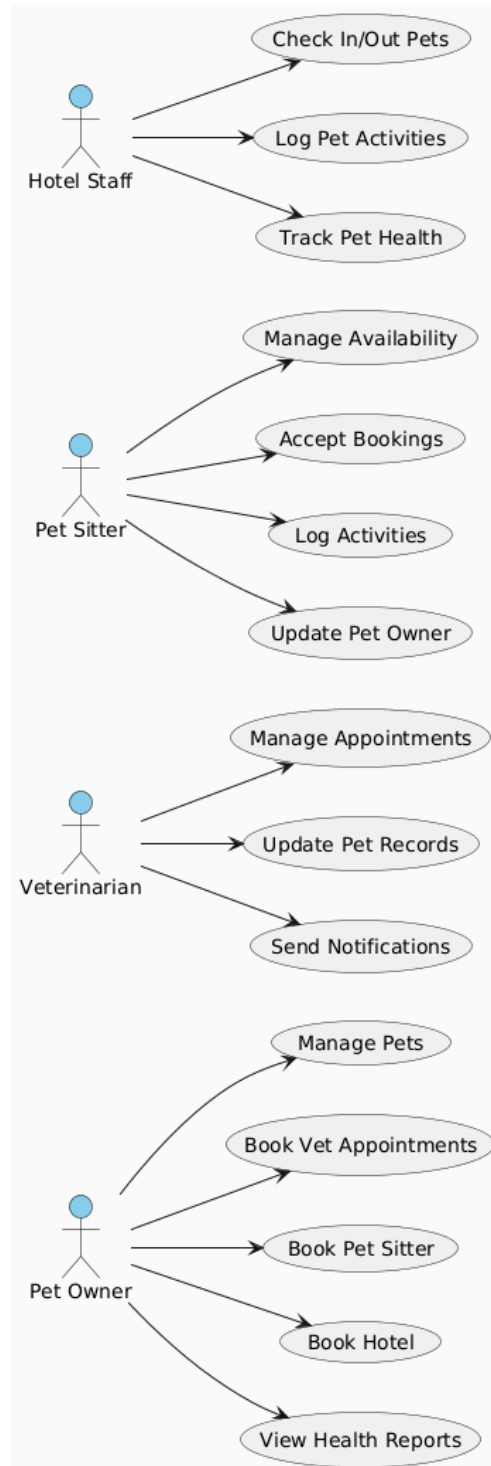
- Wants to reduce administrative overhead
- Prefers automated appointment reminders

Persona 3: Maria Garcia - Pet Sitter

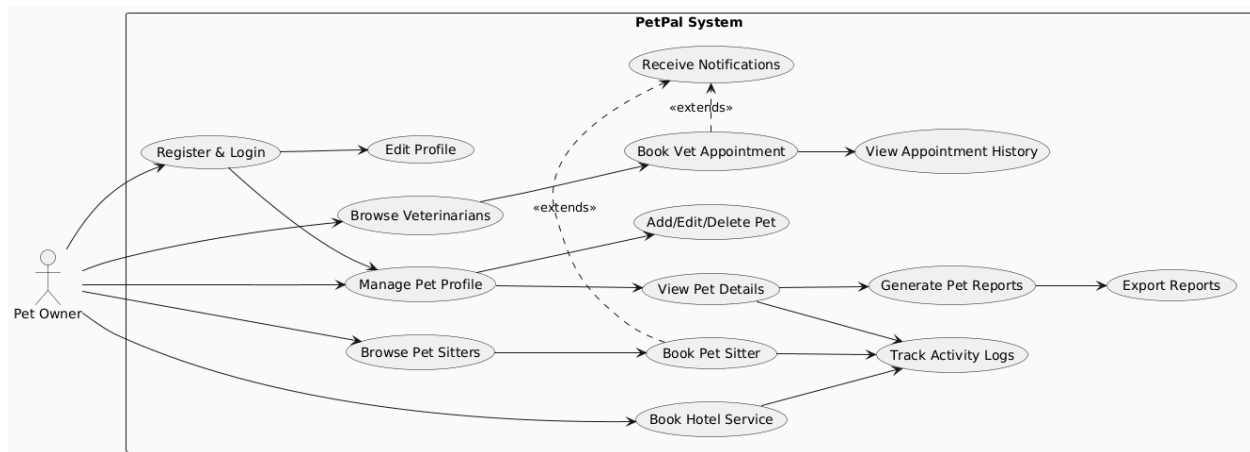
- Age: 28, independent pet care professional
- Services 5-10 clients per day
- Needs reliable booking and scheduling system
- Wants to build client trust through activity updates
- Limited technical background but willing to learn

2.0 UML Diagrams

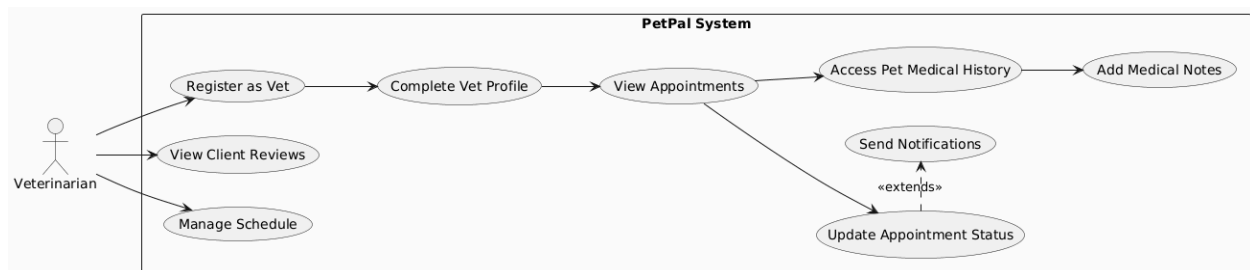
2.1 Actor/Stakeholder Diagram



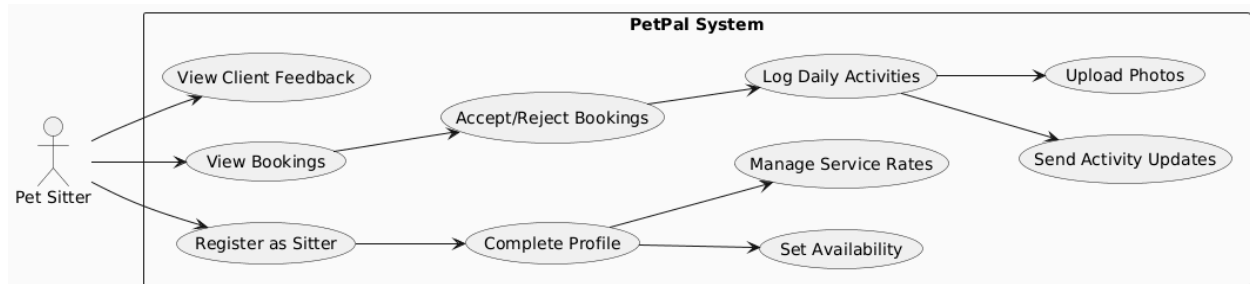
2.2 Use Case Diagram - Pet Owner



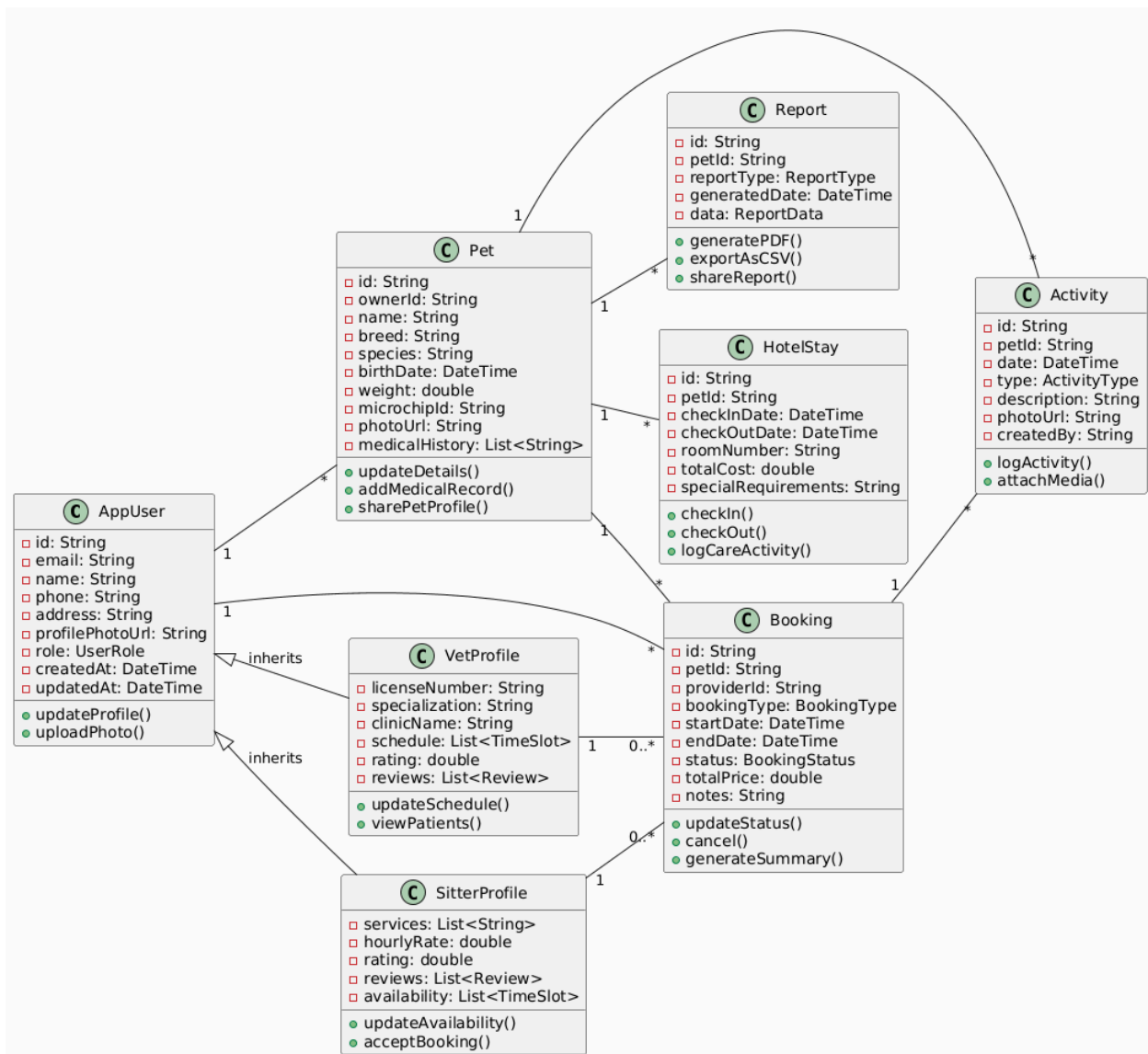
2.3 Use Case Diagram - Veterinarian



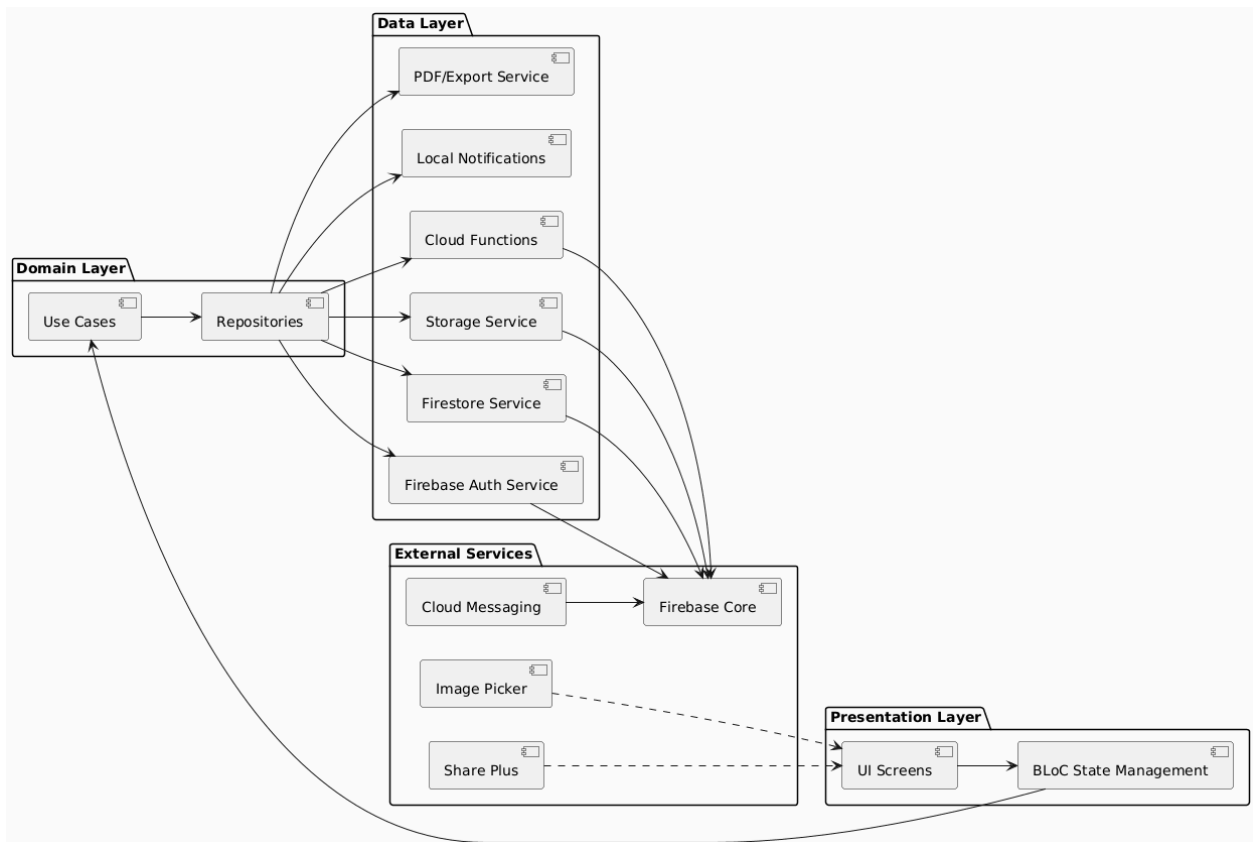
2.4 Use Case Diagram - Pet Sitter



2.5 Domain Model - Class Diagram

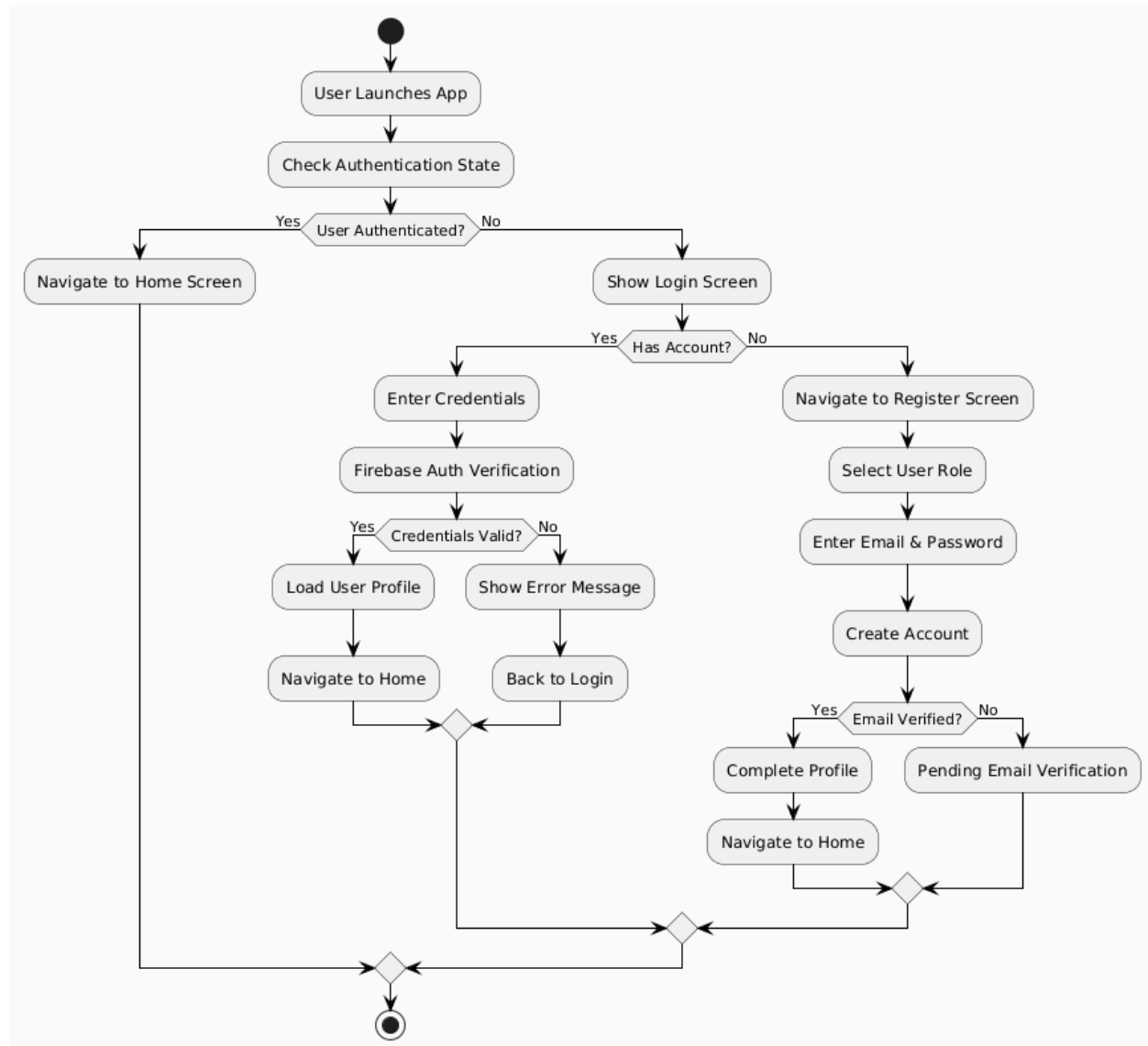


2.6 System Architecture Diagram

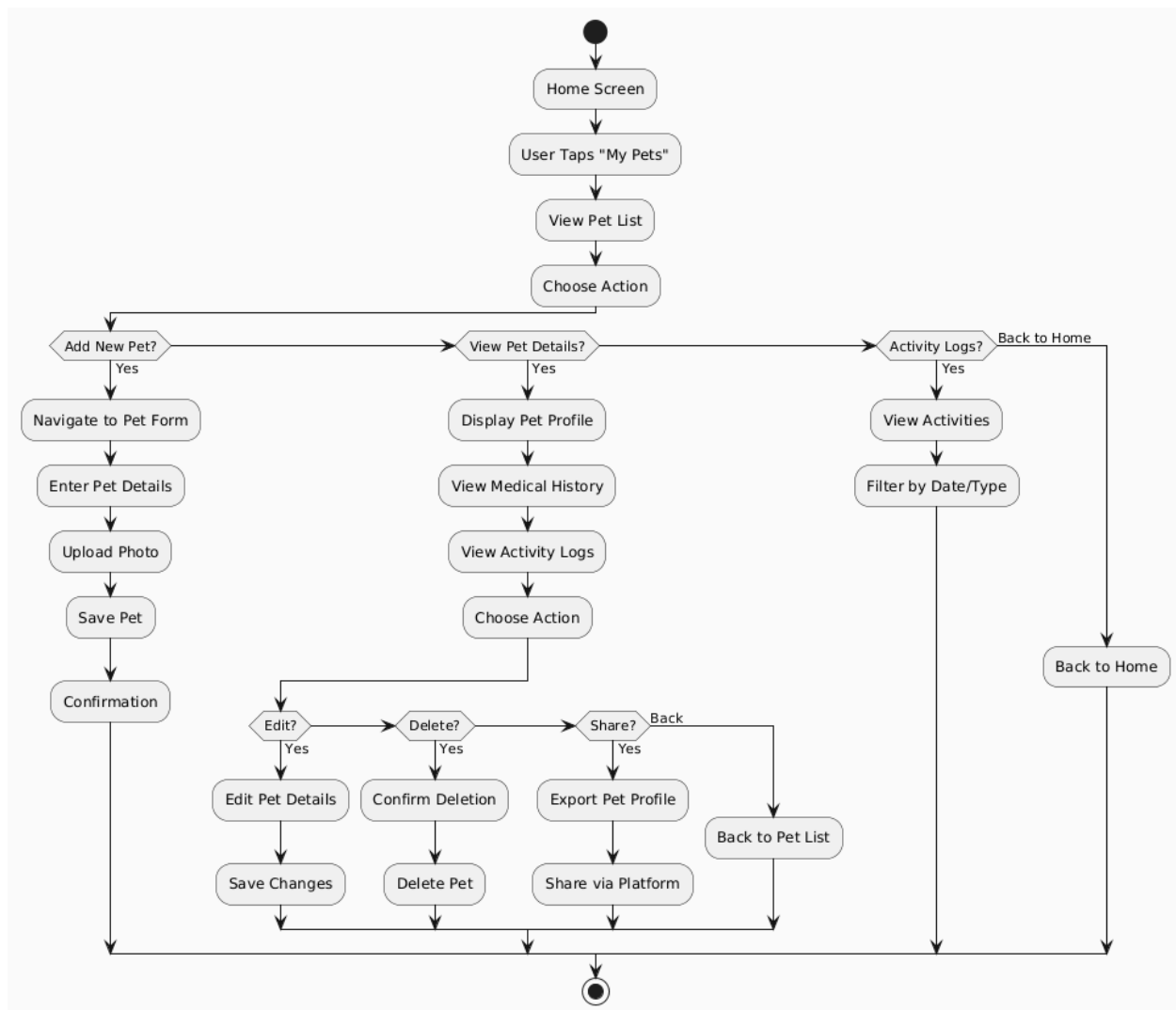


3.0 System Flow and Navigation Structure

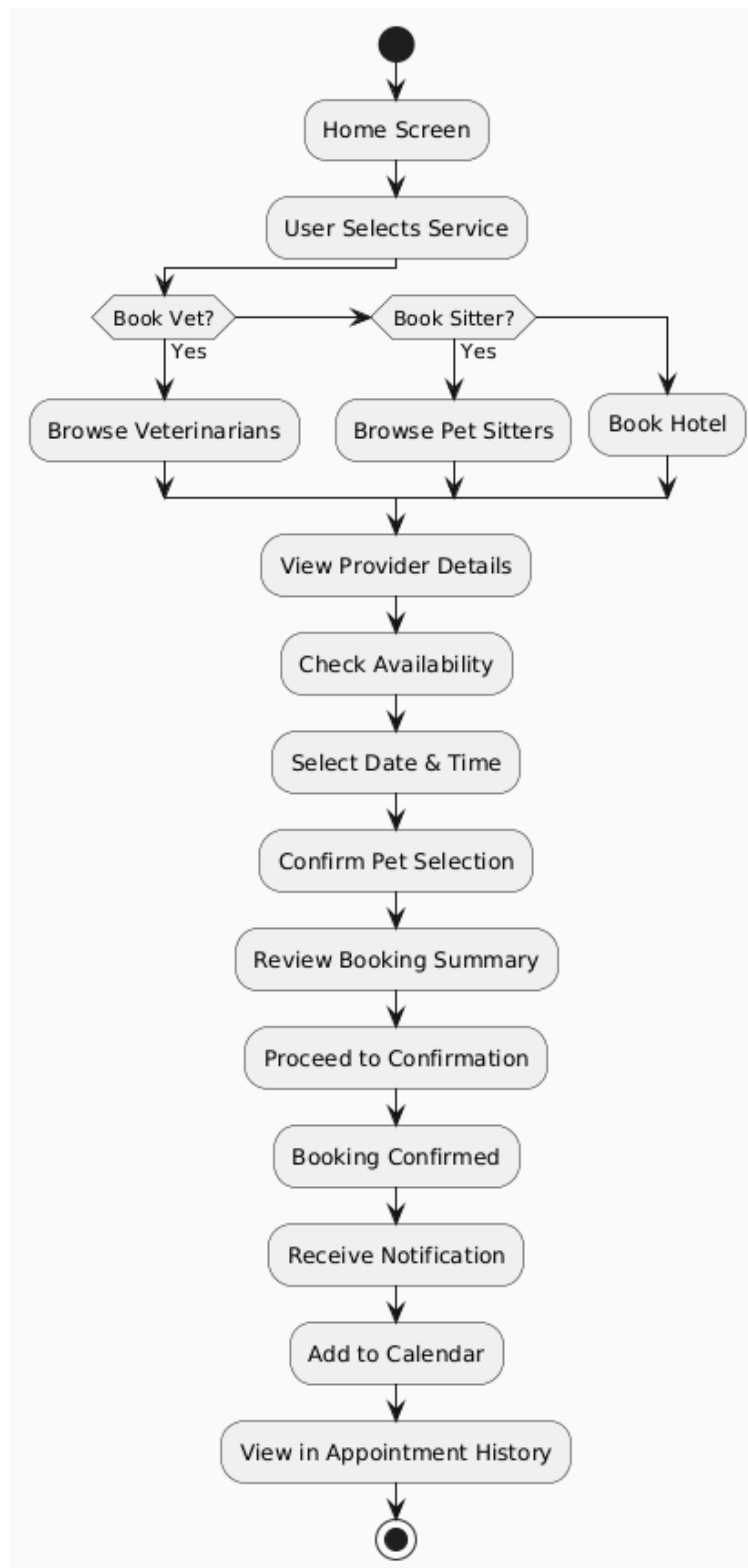
3.1 Authentication Flow



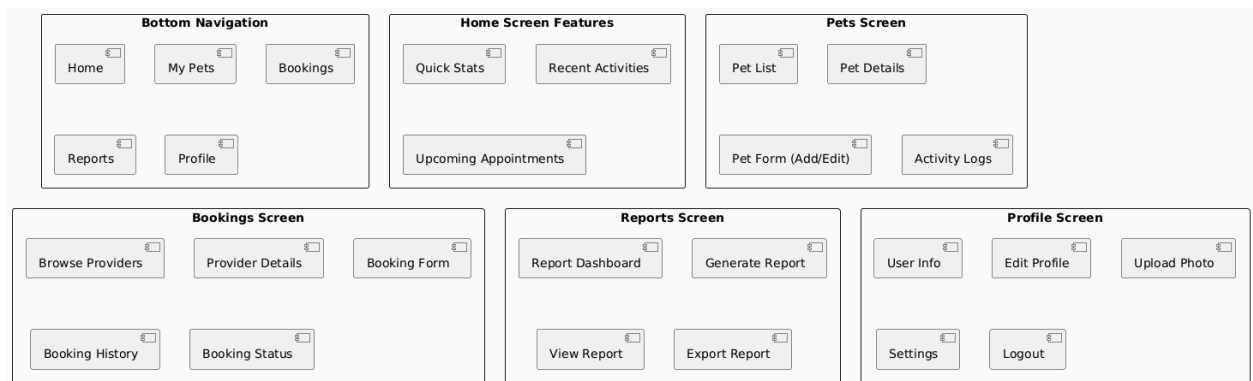
3.2 Pet Management Flow



3.3 Booking Flow (Vet/Sitter)



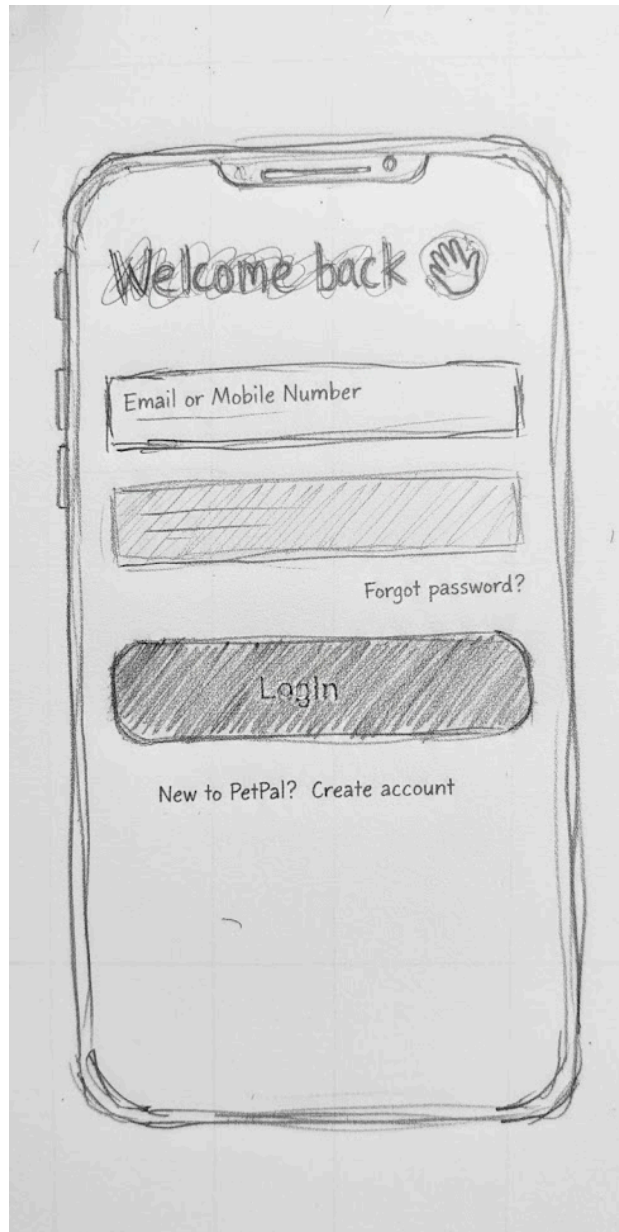
3.4 Navigation Structure



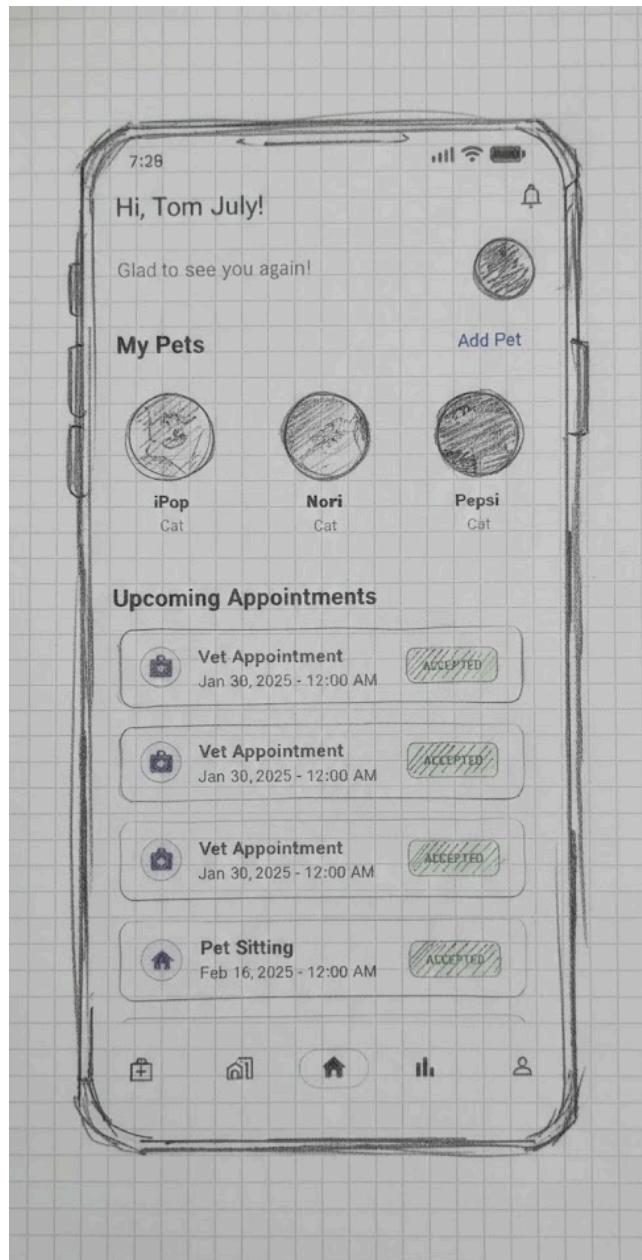
4.0 User Interface Design

4.1 Low-Fidelity Wireframes

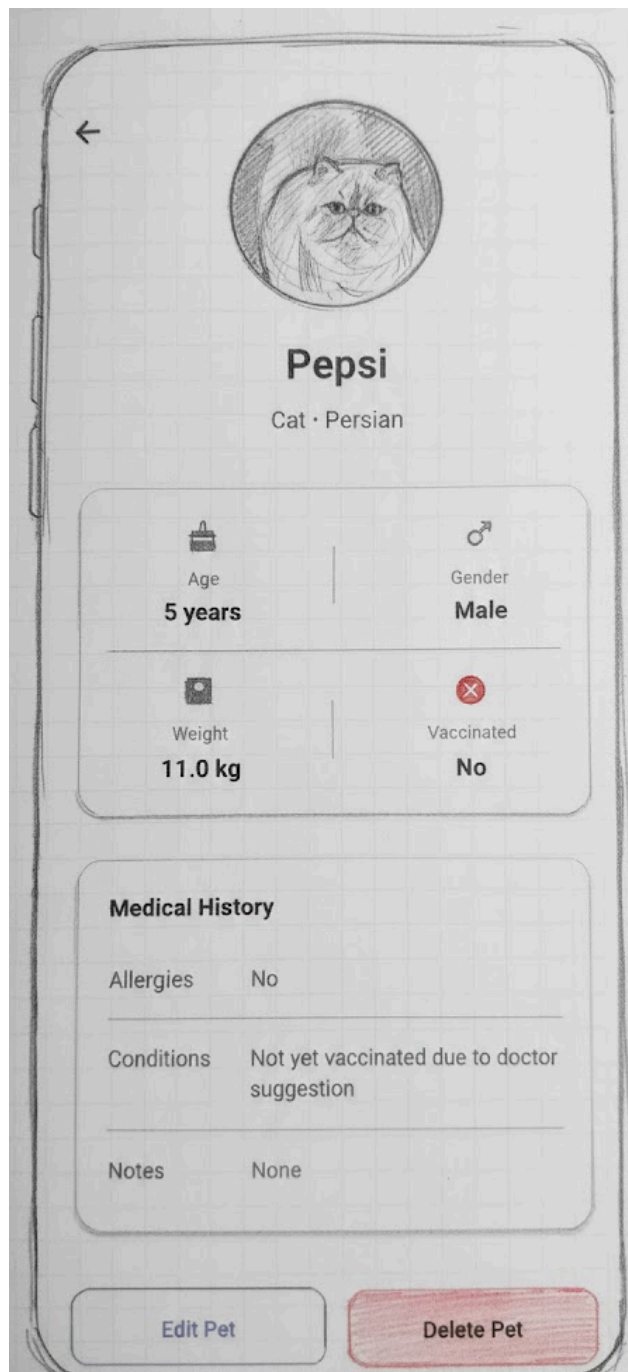
4.1.1 Login Screen Wireframe



4.1.2 Home Screen Wireframe



4.1.3 Pet List Screen Wireframe



4.1.4 Booking Form Wireframe

A hand-drawn wireframe of a mobile application interface for booking a Mac Holler. The interface is displayed on a smartphone screen with a grid background. At the top, the status bar shows the time 8:03, signal strength, Wi-Fi, and battery icons. Below the status bar is a back arrow and the title "Book Mac Holler". The form contains the following elements:

- Select pet:** A dropdown menu with "Toby" selected.
- Date:** A date picker showing "2024-01-24" with a calendar icon.
- Start:** A time picker showing "2:03 PM" with a clock icon.
- End:** A time picker showing "3:35 PM" with a clock icon.
- Notes:** A text input field containing "Has cancer".
- Confirm booking:** A large blue button at the bottom.

4.2 High-Fidelity UI Designs

4.2.1 Design System Overview

Color Palette:

- Primary Color: #2E7D32 (Emerald Green) - Trust, growth, pet care
- Secondary Color: #FF6B35 (Warm Orange) - Energy, friendliness, action
- Accent Colors: #FFC300 (Gold), #00A8FF (Sky Blue)
- Neutral: #F5F5F5 (Light Gray), #333333 (Dark Gray), #FFFFFF (White)

Typography:

- Headlines: Google Fonts - Poppins Bold (24px-32px)
- Body Text: Google Fonts - Roboto (14px-16px)
- Buttons: Poppins SemiBold (16px)

UI Components:

- Button Style: Material 3 filled buttons with rounded corners (12px radius)
- Cards: Elevation with rounded corners (16px radius), subtle shadows
- Bottom Navigation: 5 primary tabs with icons + labels
- Input Fields: Outlined style with Material 3 design

Layout Structure:

- Header: App logo, notification icon, user avatar (top bar)
- Welcome banner with gradient background (Emerald to lighter green)
- Quick statistics cards in 3-column grid layout
- Recent activities list with pet photos
- Action buttons (Book Vet, Book Sitter, View Pets)
- Bottom navigation bar

Interactive Elements:

- Notification bell shows unread count
- User avatar opens profile menu
- Statistics cards are tappable for detailed views
- Activity items show status indicators
- CTA buttons with orange accent

4.2.2 Create Account Page

6:02

← Create Account

I am a...
Pet Owner

Personal Information

First Name Last Name

Email

Password

☐ At least 6 characters

Confirm Password

☐ Passwords match

Age Date of Birth

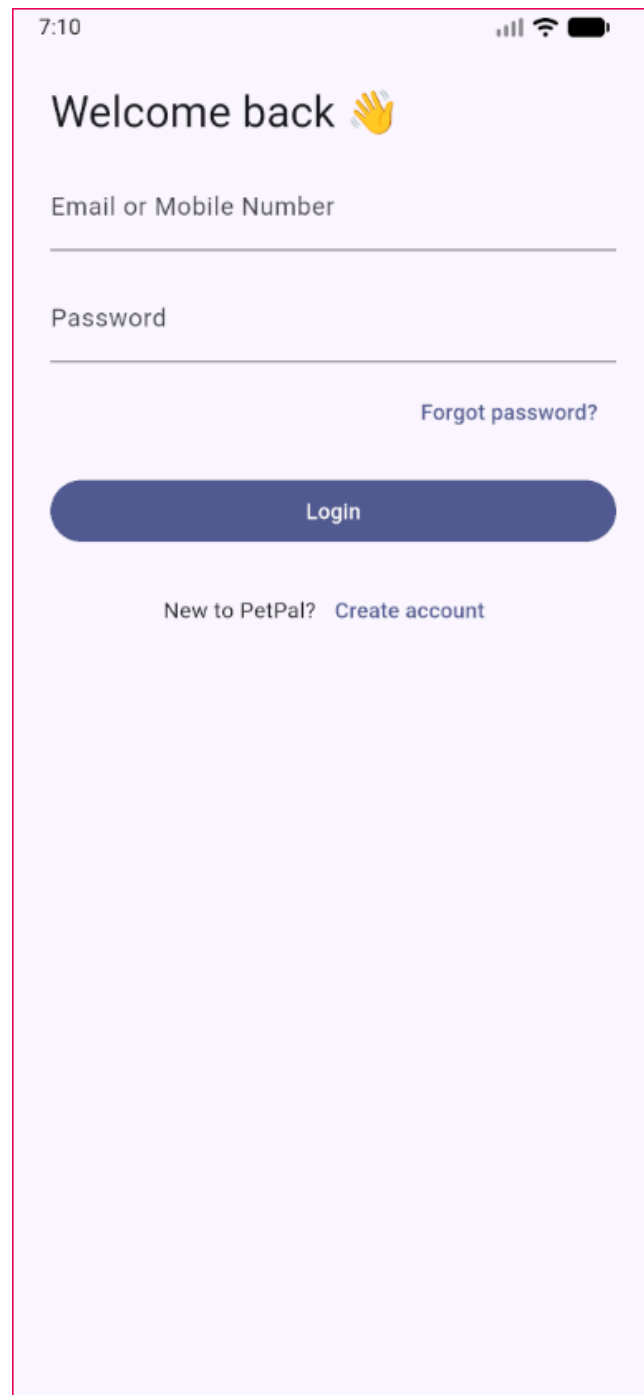
Code +60 Mobile Number

Address

Address Line 1

Address Line 2 (Optional)

4.2.3 Login Page



A mobile application login screen with a light purple background. At the top, the status bar shows the time 7:10, signal strength, Wi-Fi, and battery icons. The main heading is "Welcome back" followed by a yellow hand emoji. Below this are two input fields: "Email or Mobile Number" and "Password", each with a horizontal line underneath. To the right of the password field is a link "Forgot password?". A dark blue rounded rectangular button with the text "Login" is centered below the input fields. At the bottom, the text "New to PetPal?" is followed by a link "Create account".

7:10

Welcome back 🖐️

Email or Mobile Number

Password

[Forgot password?](#)

Login

New to PetPal? [Create account](#)

4.2.4 Reset Password Page



A mobile application screen for resetting a password. The status bar at the top shows the time 7:10, signal strength, Wi-Fi, and battery. The page has a light purple background. At the top, there is a back arrow and the title "Reset password". Below this is a text input field labeled "Email". Underneath the input field is a dark blue button with the text "Send reset link".

7:10

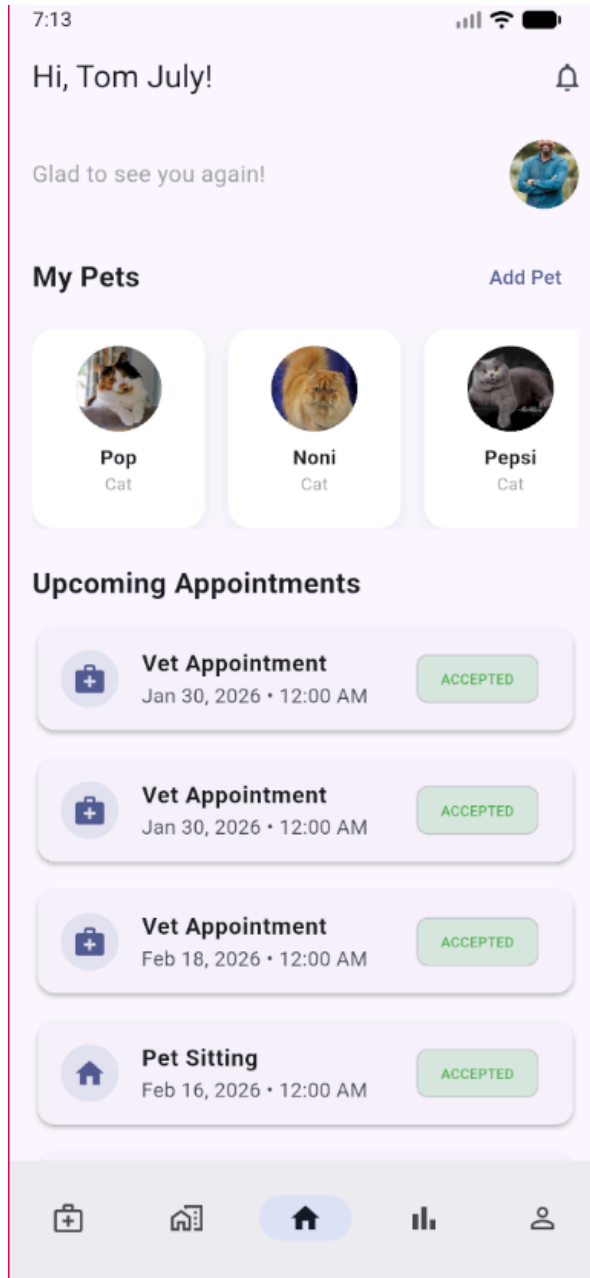
← Reset password

Email

Send reset link

4.2.5 Pet Owner Module

4.2.5.1 Pet Owner Home



4.2.5.2 Add Pet

7:17

←

Add pet

Name

Species

Breed

Age (years)

Gender

Female

▼

Weight (kg)

Vaccinated?

Medical Information

Allergies (Optional)

Medical Conditions (Optional)

Medical Notes (Optional)

Save

4.2.5.3 Pet Details

5:56





Pepsi

Cat • Persian



Age

5 years



Gender

Male



Weight

11.0 kg



Vaccinated

No

Medical History

Allergies	No
Conditions	Not yet vaccinated due to doctor suggestion
Notes	None

Edit Pet

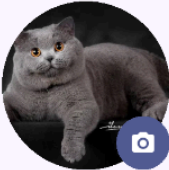
Delete Pet

4.2.5.4 Edit Pet Details

7:18

←

Edit pet



Name

Pop

Species

Cat

Breed

Calico

Age (years)

32

Gender

Female

▼

Weight (kg)

8.0

Vaccinated?

☒

Medical Information

Allergies (Optional)

No

Medical Conditions (Optional)

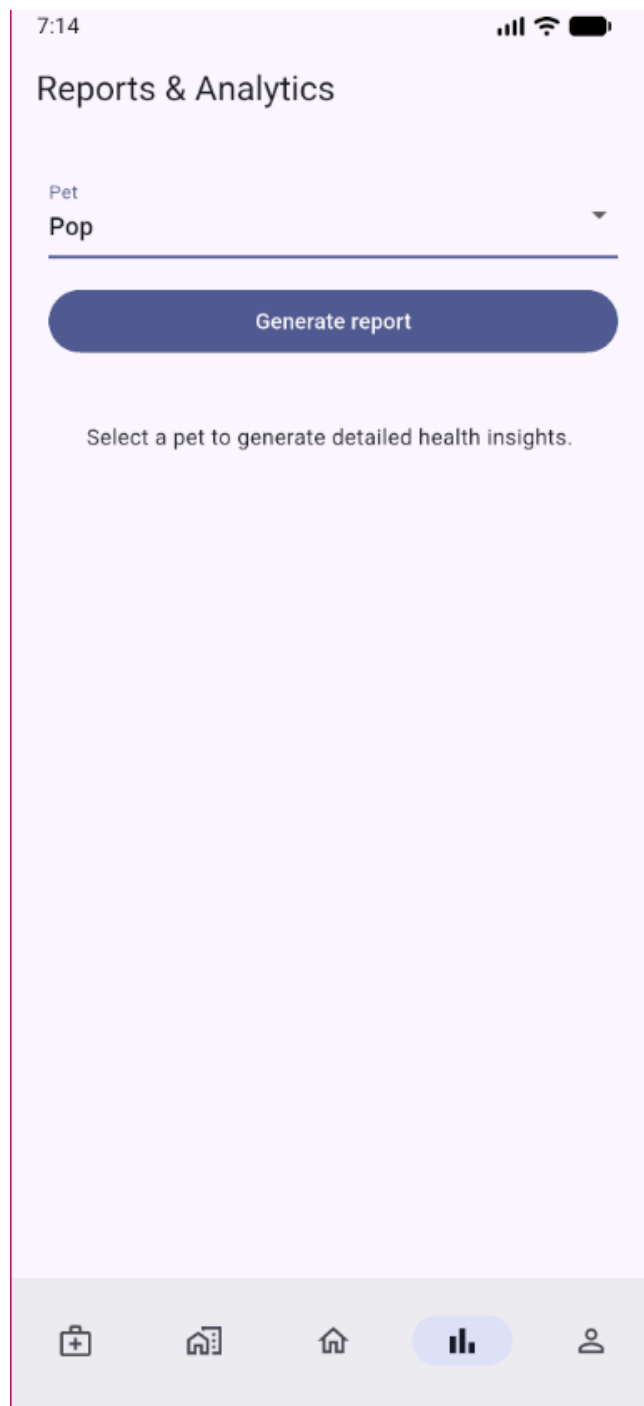
Fever

Medical Notes (Optional)

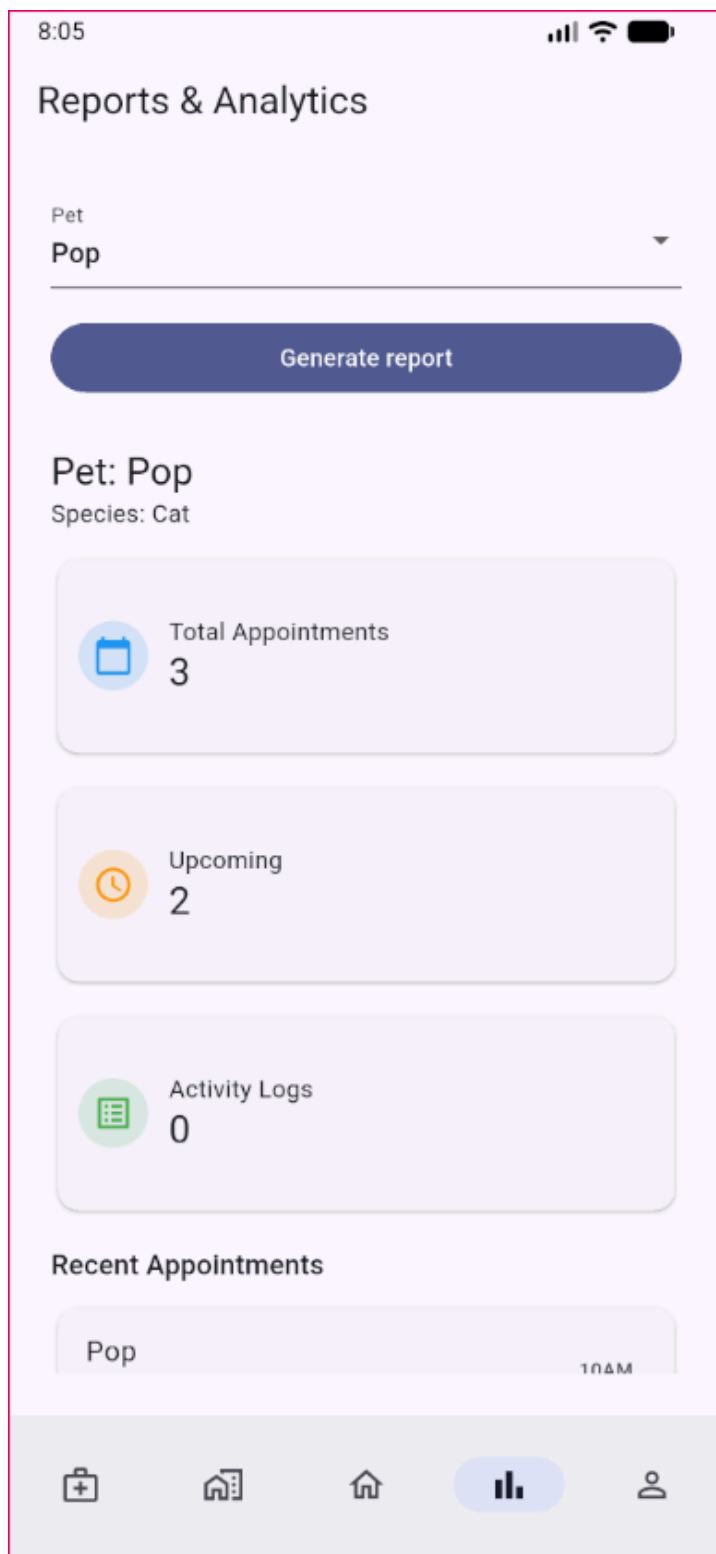
No

Save

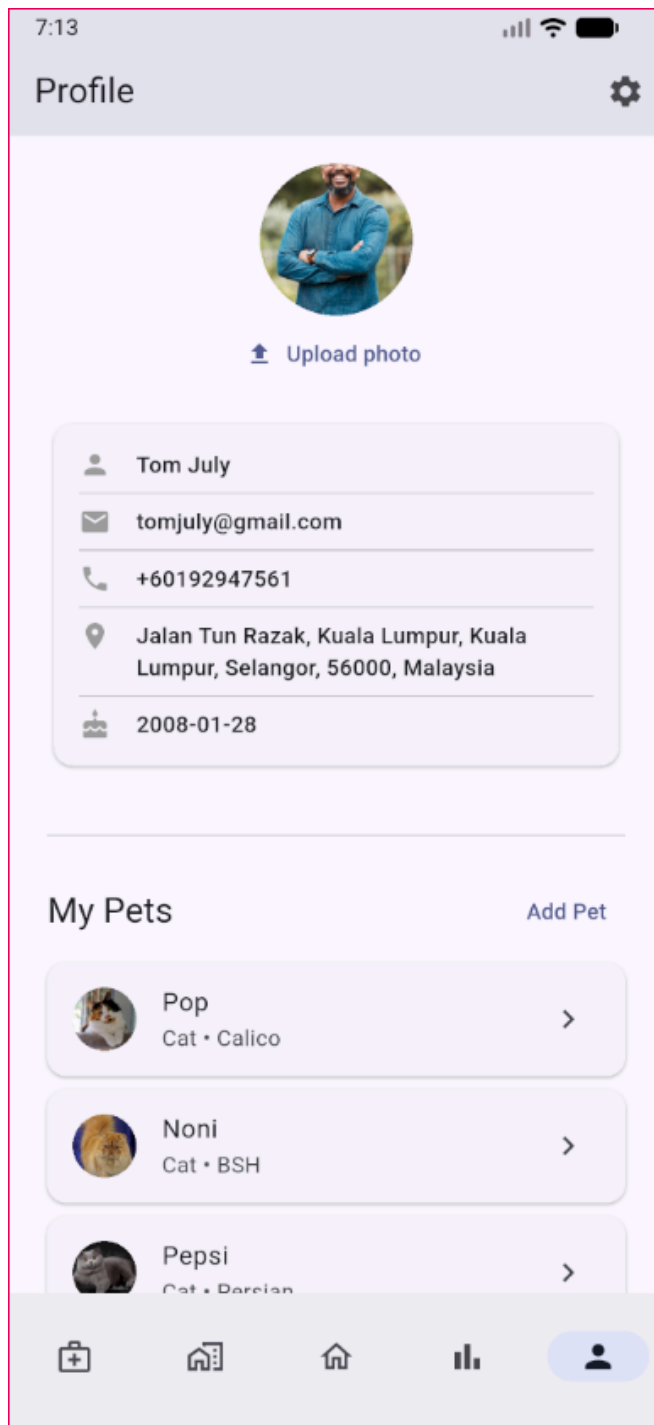
4.2.5.5 Pet Report



4.2.5.6 Pet Report Details



4.2.5.7 Pet Owner Profile



4.2.5.8 Pet Owner Edit Profile

7:17

← Edit Profile

Personal Information

Full name
Tom July

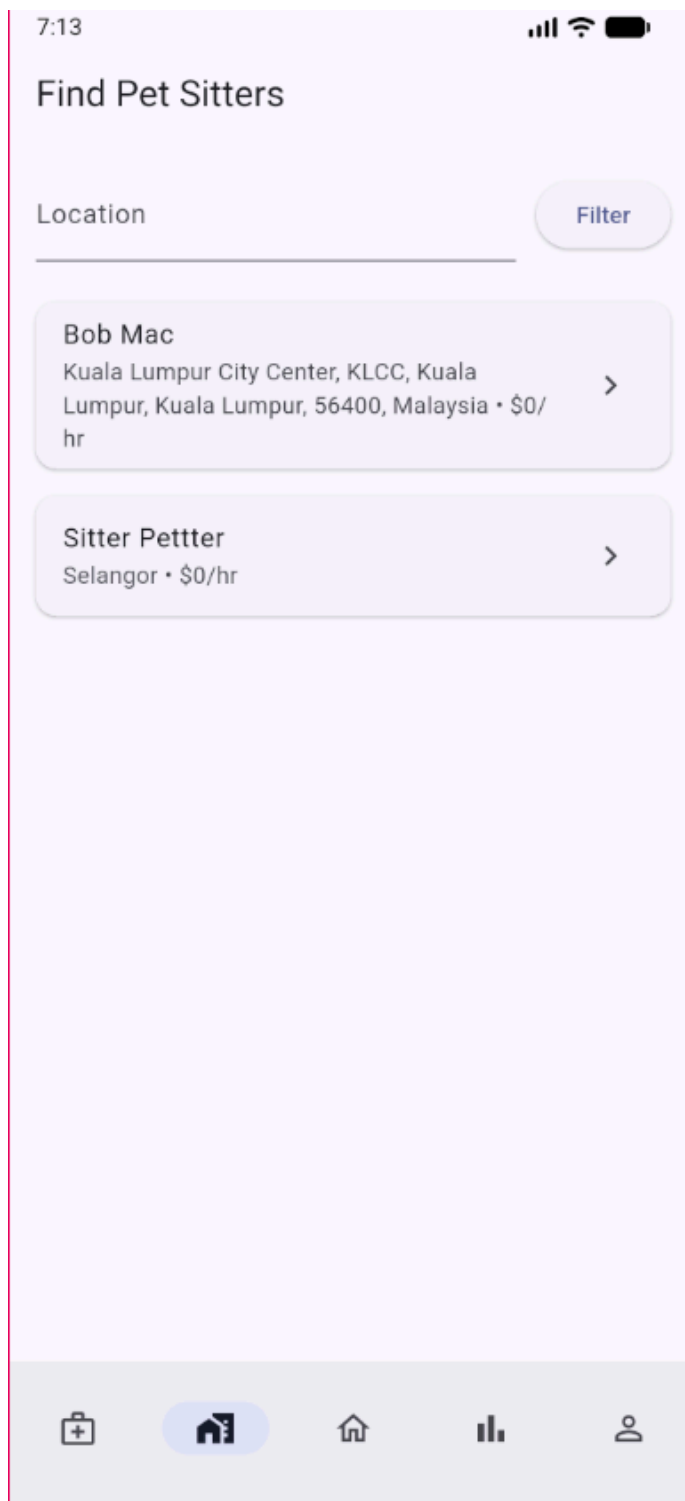
Phone
+60192947561

Address
Jalan Tun Razak, Kuala Lumpur, Kuala Lumpur, S

Birthday (YYYY-MM-DD)
2008-01-28

Save Changes

4.2.5.9 Pet Sitter Booking



4.2.5.10 Pet Sitter Booking Details

7:16

←

Bob Mac

BM

Bob Mac

Kuala Lumpur City Center, KLCC, Kuala Lumpur,
Kuala Lumpur, 56400, Malaysia

10 yrs experience **\$0/hr**

Services

Day Care

Overnight Sitting

Home Visit

Pet Types Accepted

Cat

Rabbit

Bird

About

Has 10+ years experience in pet sitter business

Availability

Mon

Fri

Tue

Wed

Thu

start: 07:00

end: 19:00

Book sitter

Contact

4.2.5.11 Pet Sitter Confirm Booking

8:04

← Book Bob Mac

Select pet

Toby

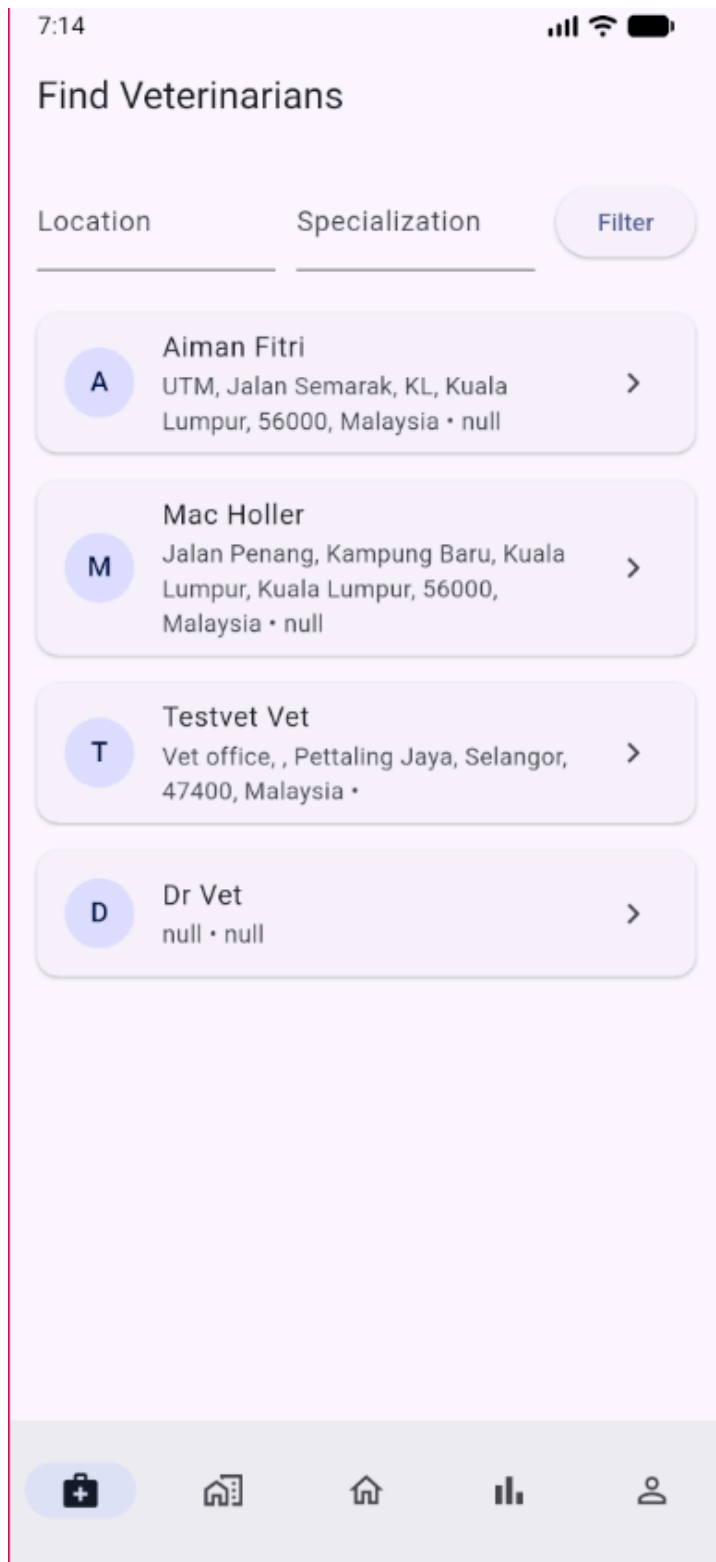
2026-01-28 - 2026-01-28

Notes for sitter

Confirm sitter booking

The screenshot shows a mobile application interface for booking a pet sitter. At the top, the status bar displays the time 8:04, signal strength, Wi-Fi, and battery icons. Below this is a header with a back arrow and the text 'Book Bob Mac'. A dropdown menu labeled 'Select pet' shows 'Toby' as the selected option. Below the dropdown is a date range '2026-01-28 - 2026-01-28' with a calendar icon to its right. Underneath is a section titled 'Notes for sitter' with a horizontal line for input. At the bottom is a large blue button with the text 'Confirm sitter booking'.

4.2.5.12 Veterinary Booking



4.2.5.13 Veterinary Booking Details

8:02



Mac Holler



Mac Holler

Jalan Penang, Kampung Baru, Kuala Lumpur,
Kuala Lumpur, 56000, Malaysia

Specialization

General Practice

Dermatology

Emergency Care

Availability

Mon, Tue, Sun, Sat, Wed, Thu, Fri | 08:00 - 20:00

Book appointment

 Contact

4.2.5.14 Veterinary Confirm Booking

The screenshot shows a mobile application interface for confirming a veterinary booking. At the top, the status bar displays the time 8:03, signal strength, Wi-Fi, and battery icons. Below the status bar is a header with a back arrow and the title "Book Mac Holler". The main form area includes a "Select pet" dropdown menu with "Toby" selected. Below this is a "Date" field showing "2026-01-24" with a calendar icon. Two time selection buttons are present: "Start 2:05 PM" and "End 3:05 PM", each with a clock icon. A "Notes" section contains the text "Has cancer". At the bottom of the form is a large blue button labeled "Confirm booking".

8:03

← Book Mac Holler

Select pet
Toby

Date
2026-01-24

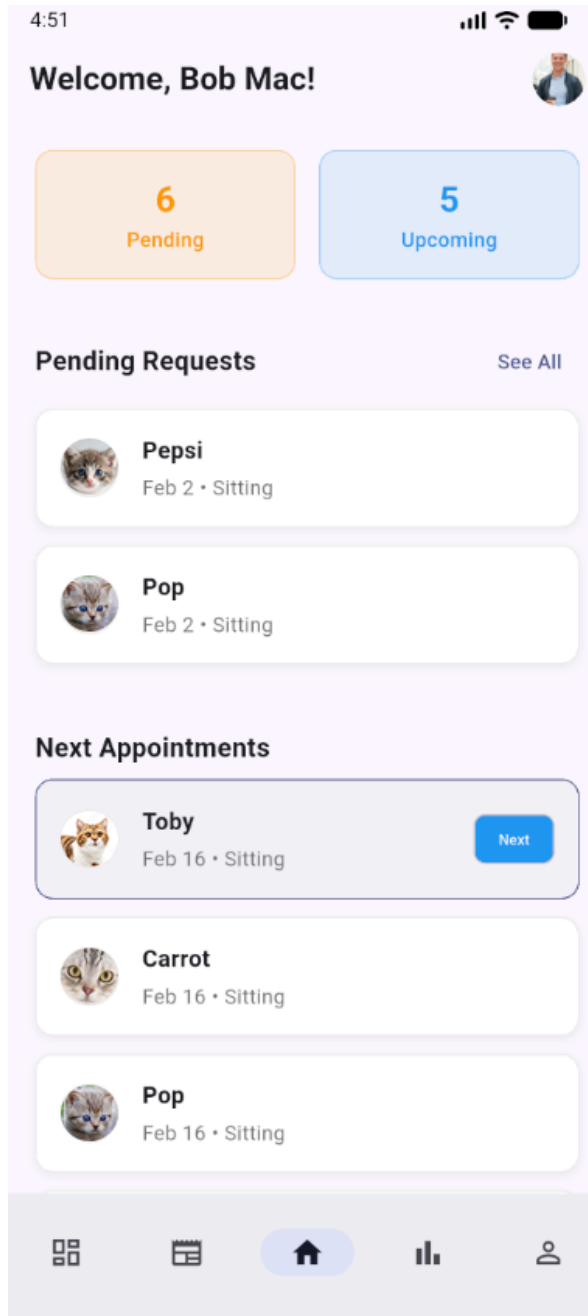
Start 2:05 PM End 3:05 PM

Notes
Has cancer

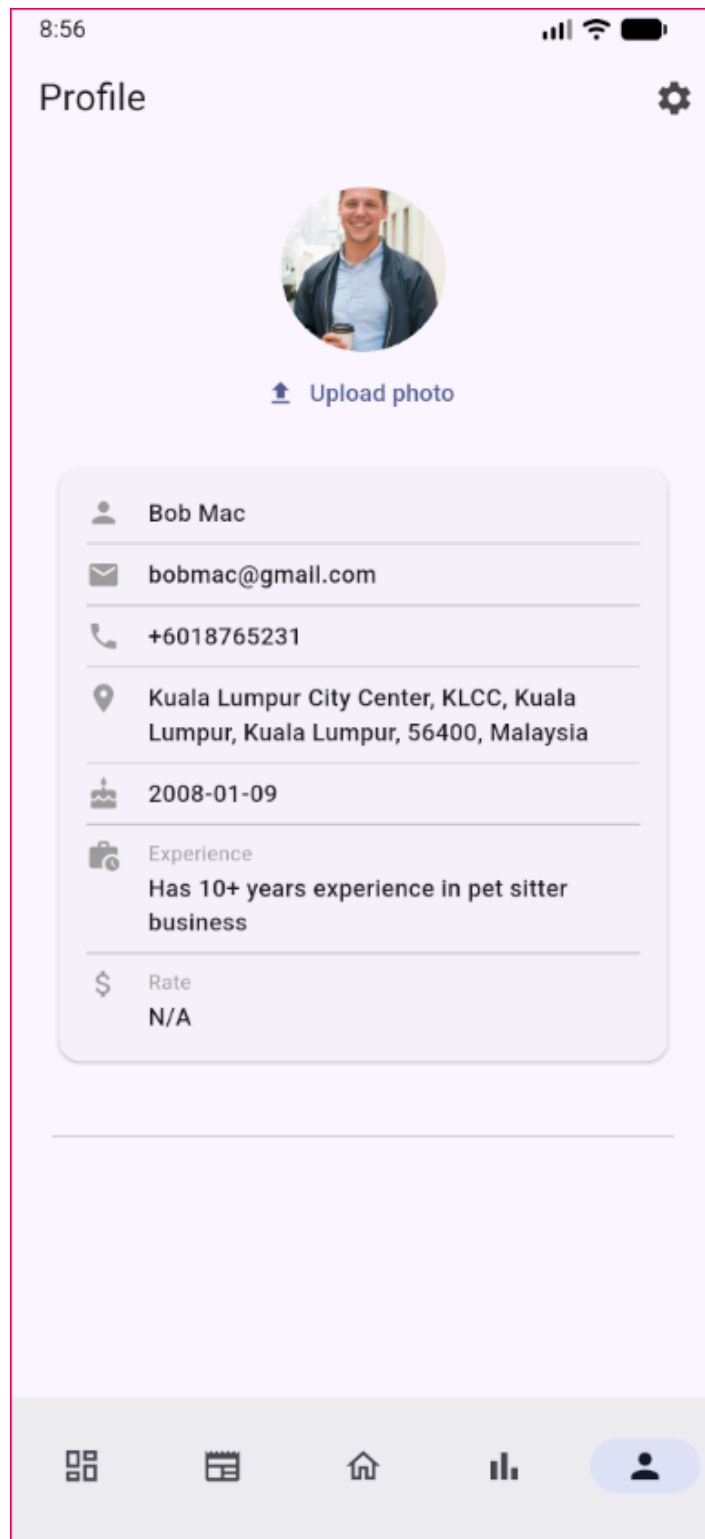
Confirm booking

4.2.6 Pet Sitter Module

4.2.6.1 Pet Sitter Homepage



4.2.6.2 Pet Sitter Profile



4.2.6.3 Pet Sitter Edit Profile

2:01

← Edit Profile

Years of Experience
10

Hourly Rate (RM)
10.0

Service Area (State)
Kuala Lumpur

Pet Types Accepted

☐ Dog ☒ Cat ☒ Rabbit ☒ Bird

☐ Other

Services Provided

☒ Day Care ☒ Overnight Sitting

☒ Home Visit ☐ Boarding

Work Availability

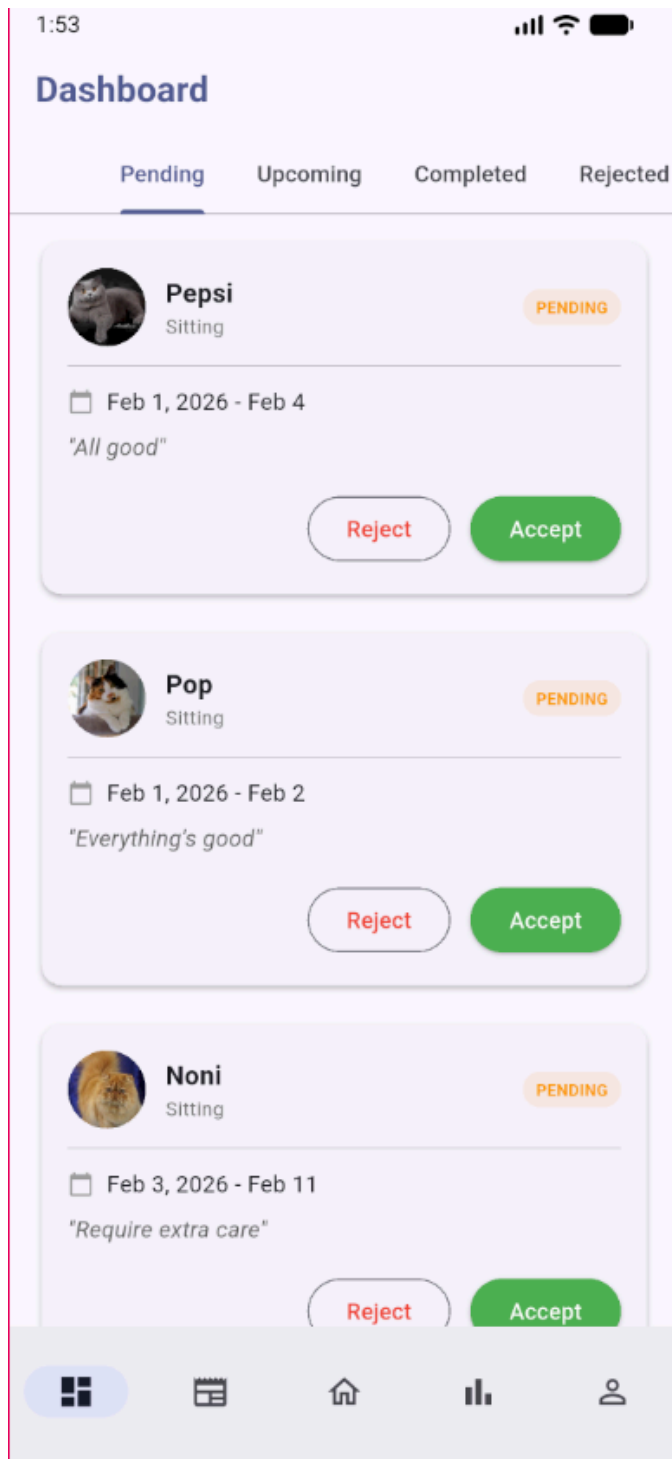
☒ Mon ☒ Tue ☒ Wed ☒ Thu

☒ Fri ☐ Sat ☐ Sun

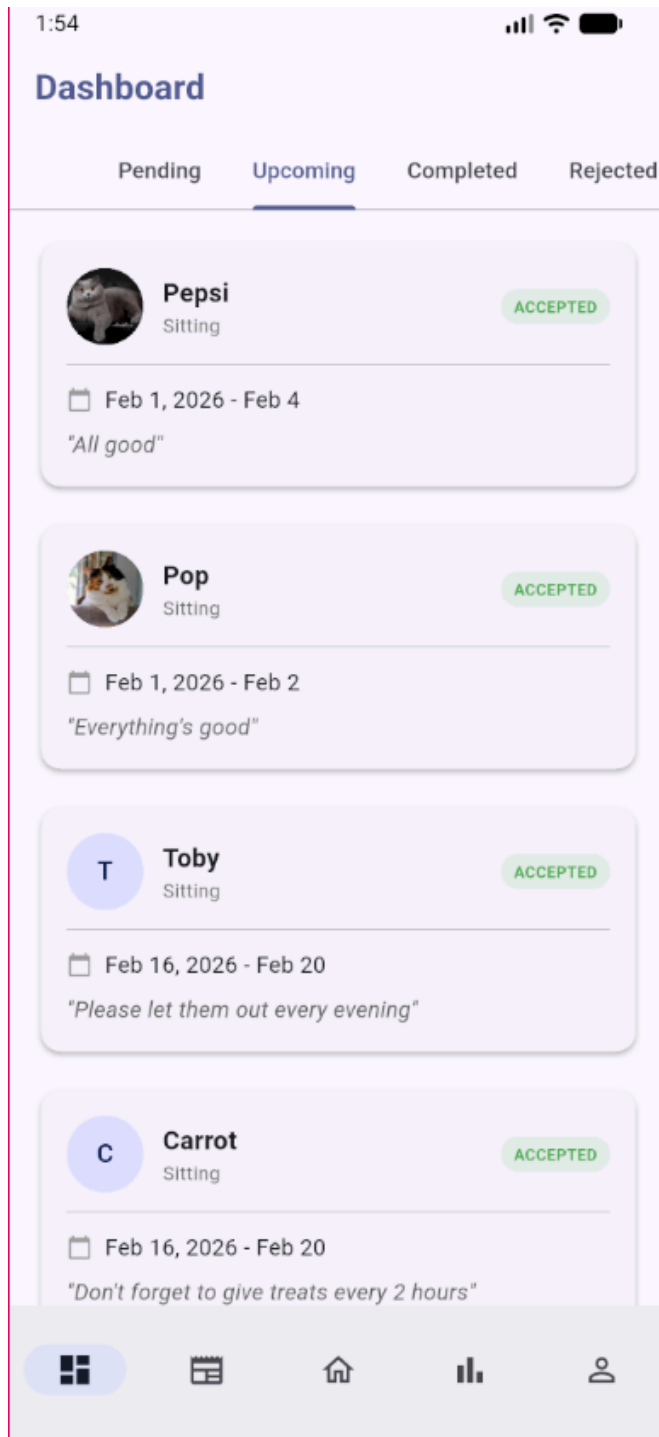
7:00 AM 7:00 PM

Save Changes

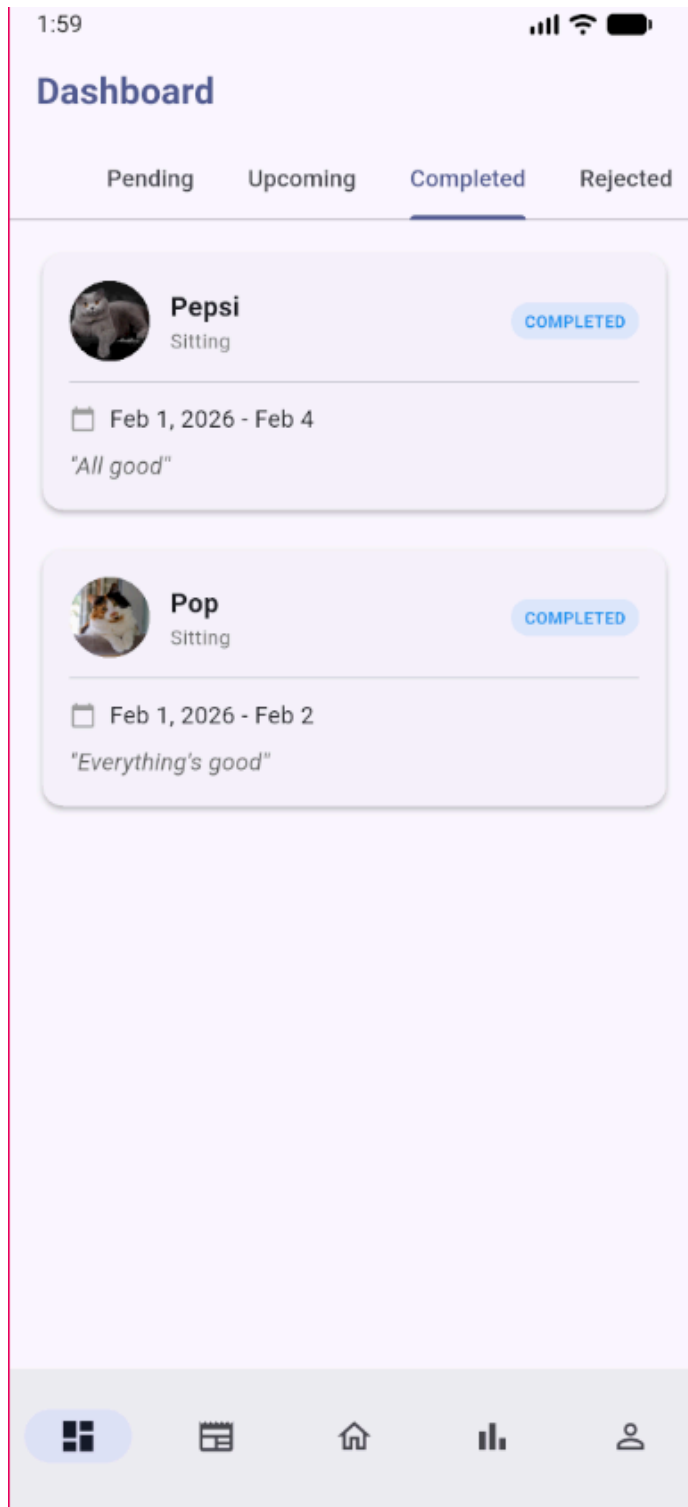
4.2.6.4 Pet Sitter Dashboard - Pending



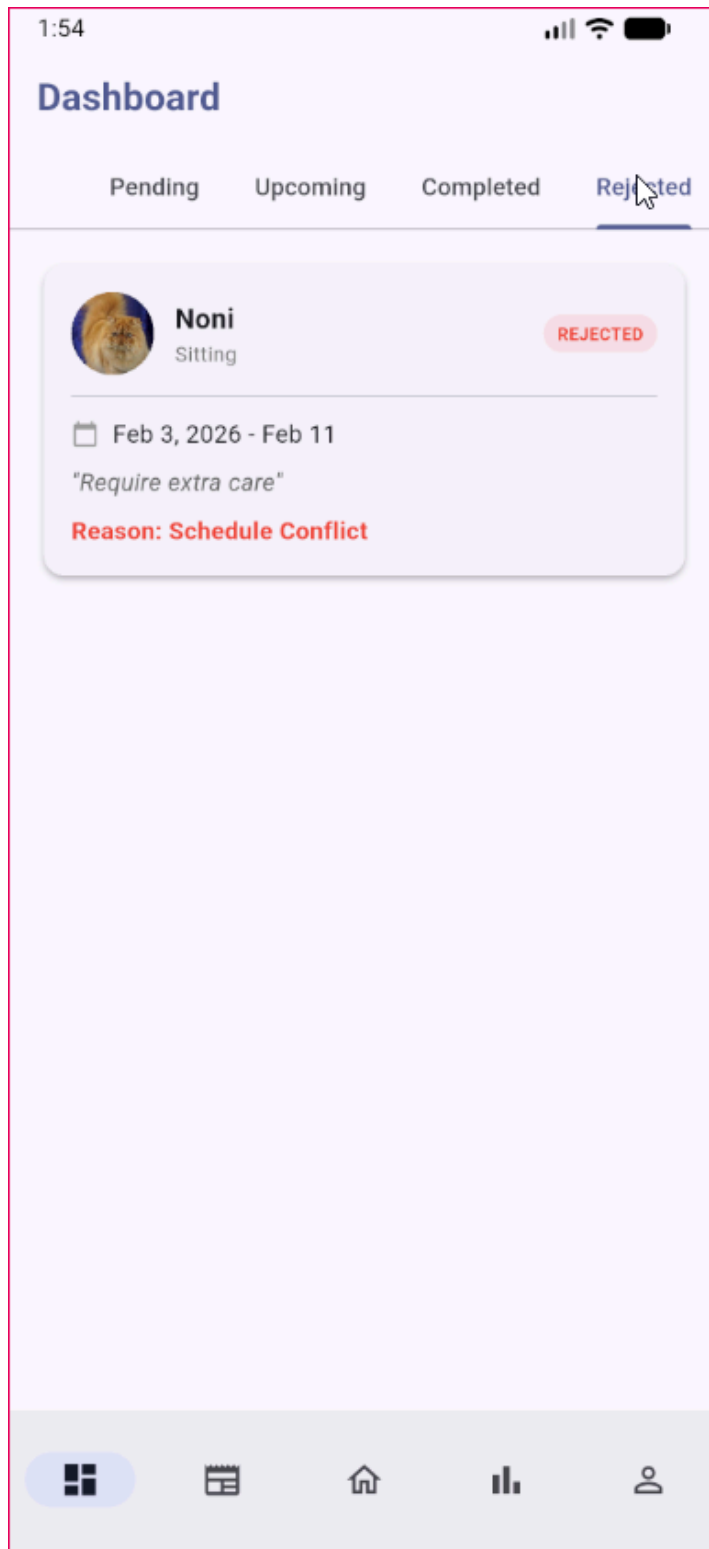
4.2.6.5 Pet Sitter Dashboard - Upcoming



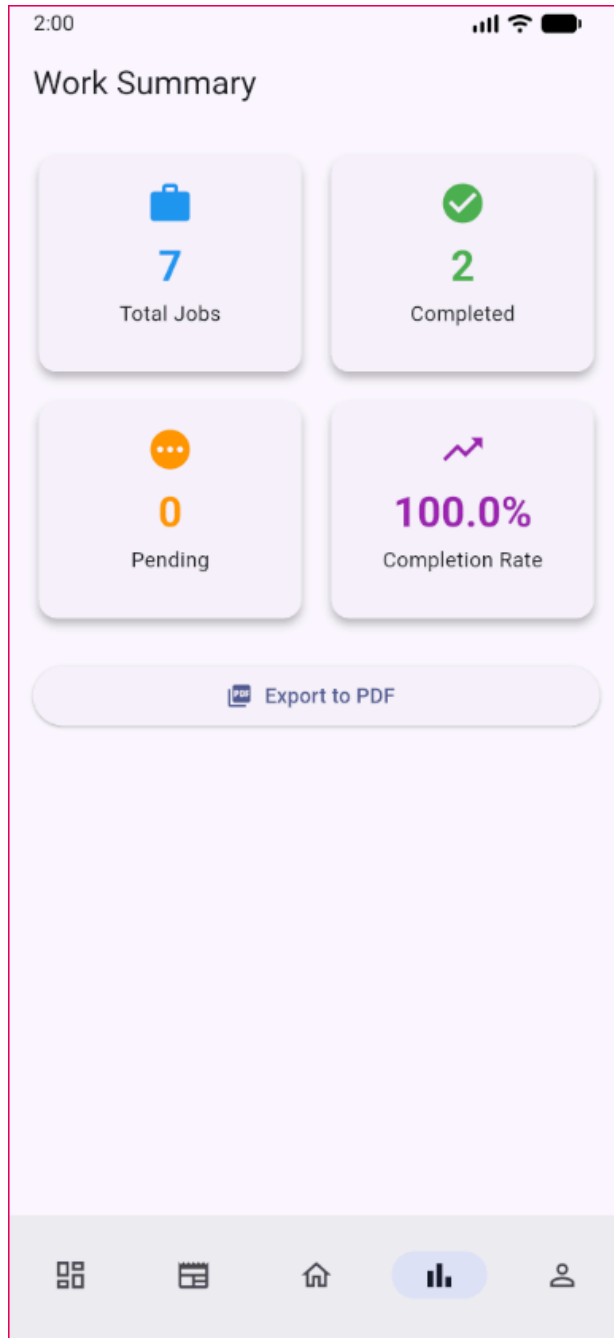
4.2.6.6 Pet Sitter Dashboard - Completed



4.2.6.7 Pet Sitter Dashboard - Rejected



4.2.6.8 Pet Sitter Report



4.2.6.9 Community Feed

8:57



Community Feed



Sitter 1

2 hours ago



Had a great time sitting for this lovely energetic husky today! #petpal #sitterlife



Like



Comment



Share



Sitter 2

2 hours ago

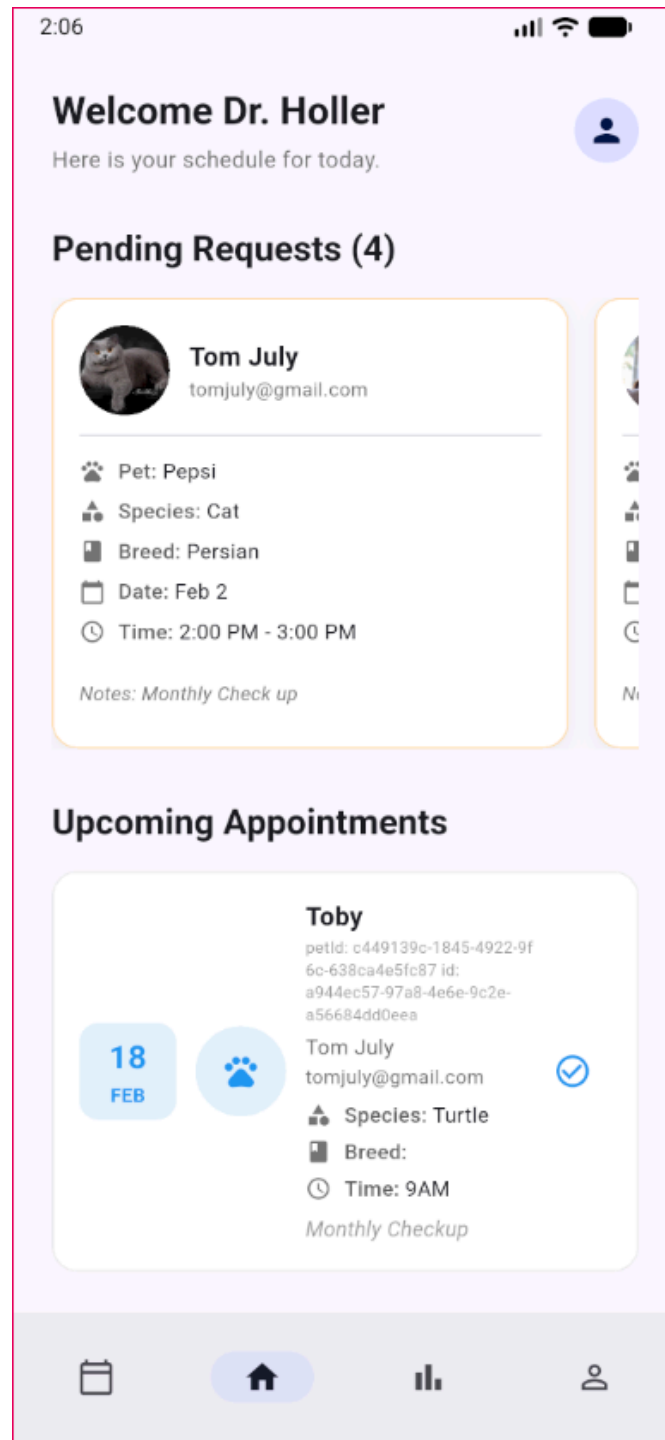


Had a great time sitting for this lovely energetic husky today! #petpal #sitterlife

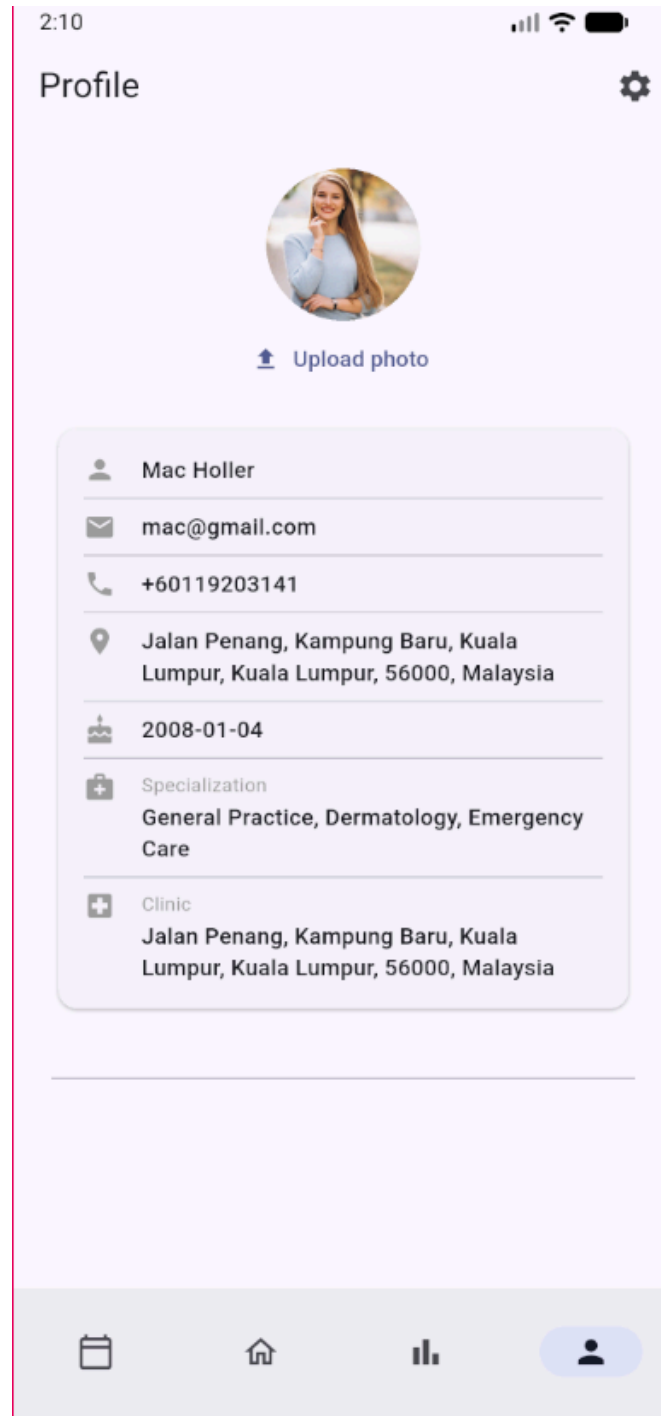


4.3 Veterinary Module

4.3.1 Veterinary Homepage



4.3.2 Veterinary Profile



4.3.3 Veterinary Edit Profile

2:10

←

Edit Profile

Personal Information

Full name

Mac Holler

Phone

+60119203141

Address

Jalan Penang, Kampung Baru, Kuala Lumpur, Ku

Jalan Penang, Kampung Baru, Kuala Lumpur, Kuala Lumpur, 56000, Malaysia

Birthday (YYYY-MM-DD)

2008-01-04

Professional Details

Specialization

✓ General Practice

Surgery

✓ Dermatology

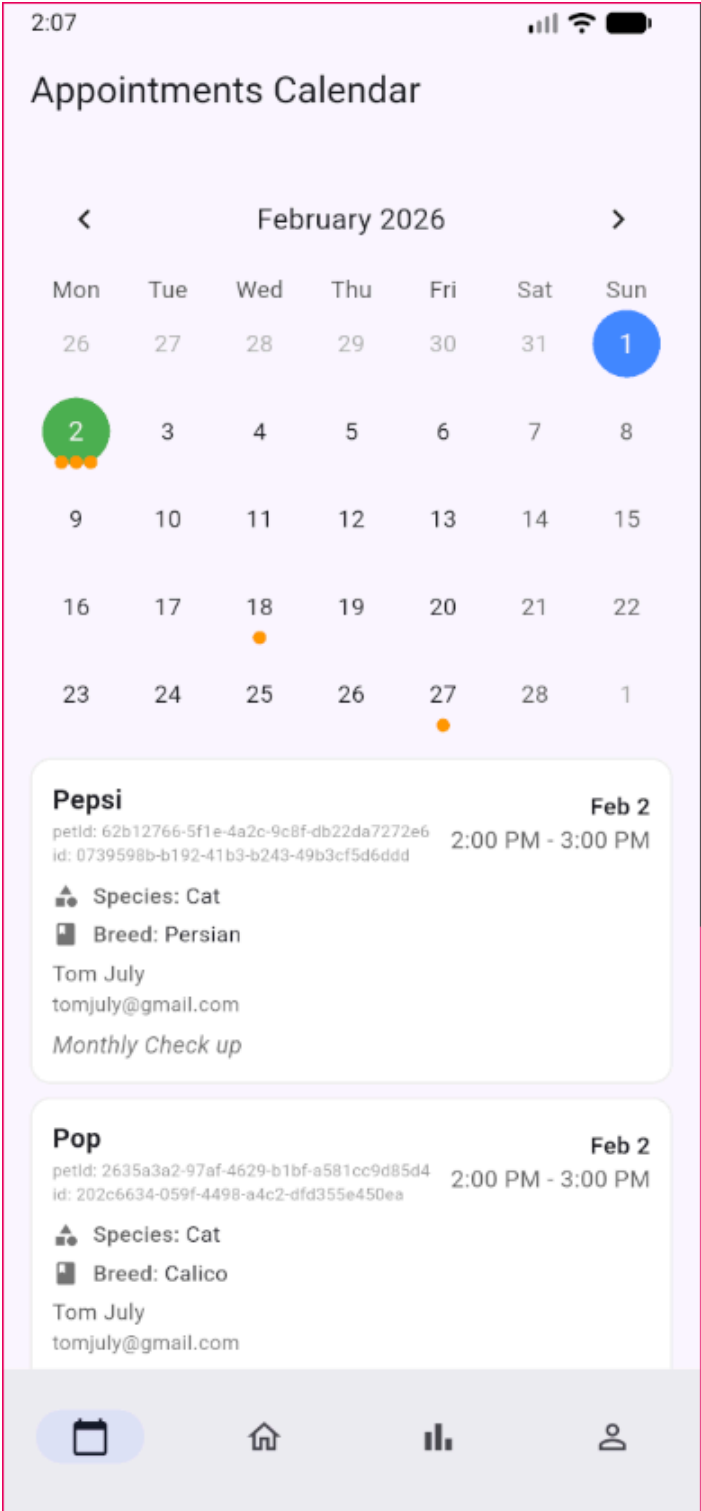
Exotic Animals

✓ Emergency Care

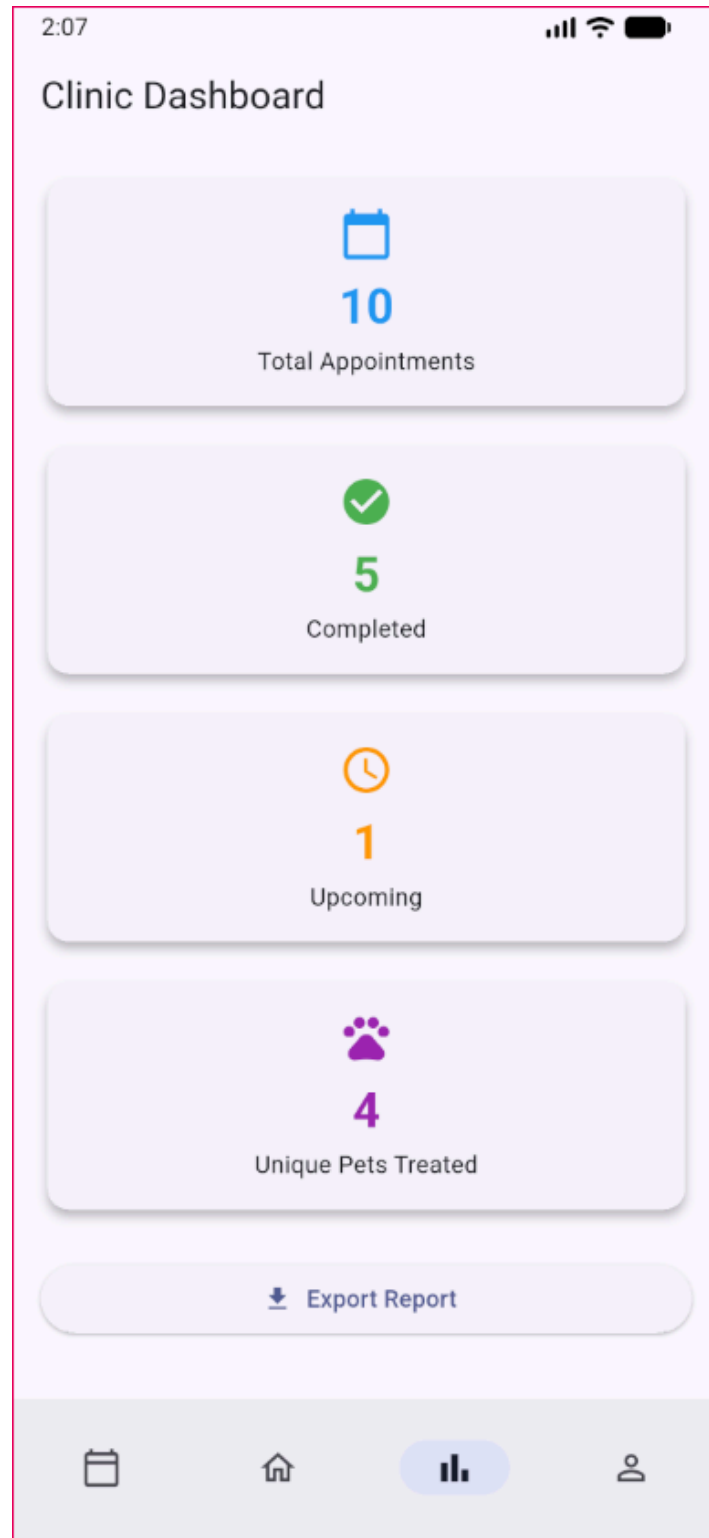
Dentistry

Internal Medicine

4.3.4 Veterinary Calendar



4.3.5 Veterinary Report



5.0 How the Proposed Design Meets User Needs

5.1 Pet Owner Needs & Solutions

User Need	Design Solution	Benefit
Centralized Pet Health Records	Comprehensive pet profiles with medical history, vaccination records, and activity logs	Users no longer need to maintain scattered records across vet offices and email
Easy Appointment Scheduling	Integrated booking system with calendar view and automated reminders	Reduces time spent on phone calls and reduces missed appointments
Trustworthy Care Provider Selection	Provider browsing with ratings, reviews, specializations, and availability	Builds confidence in service quality and provider reliability
Activity Tracking	Real-time activity logs with photo uploads from sitters/vets	Peace of mind when pet is away; comprehensive activity history
Health Insights	Automated report generation (health trends, vaccination tracking)	Enables proactive health management and informed veterinary discussions
Multi-Pet Management	Separate profiles for each pet with shared booking and reporting	Streamlines management of multiple pets with individual needs
Service Coordination	Unified platform for vet, sitter, and hotel bookings	Single point of contact for all pet services
Emergency Access	Shareable pet profiles with exportable health records	Critical information easily accessible in emergencies

5.2 Veterinarian Needs & Solutions

User Need	Design Solution	Benefit
Patient History Access	Immediate access to complete pet medical history	Better diagnostic decisions and reduced consultation time
Appointment Management	Centralized booking system with calendar sync	Reduced double-booking and administrative overhead
Communication with Owners	In-app notifications and follow-up messaging	Improved patient care continuity and owner compliance
Service Discovery	Vet profile with specializations and availability	Attracts appropriate patients seeking specific expertise
Record Keeping	Digital medical notes and prescription tracking	Compliance with record-keeping regulations
Reputation Building	Client reviews and ratings display	Enhanced visibility and credibility

5.3 Pet Sitter Needs & Solutions

User Need	Design Solution	Benefit
Booking Management	Clear booking calendar with pet details	Never miss bookings; organized schedule
Service Showcase	Detailed sitter profile with services, rates, and photos	Attract more clients with professional presentation
Client Communication	Activity logging and photo sharing during services	Transparent communication builds trust and encourages repeat bookings
Availability Management	Easy availability calendar updates	Avoid incorrect bookings and manage multiple pet requests
Payment & Ratings	Transparent pricing and review system	Fair compensation and reputation building
Reliability Features	Confirmation notifications and pet-specific care instructions	Ensure pets receive proper care; reduce client anxiety

5.4 Hotel/Facility Needs & Solutions

User Need	Design Solution	Benefit
Check-in/Check-out Management	Streamlined digital check-in process	Reduce front desk workload and errors
Pet Health Tracking	Comprehensive activity logging during stays	Track health status and provide owner updates
Owner Communication	Real-time activity and photo updates	Increased owner confidence in facility
Special Requirements	Detailed pet profile with care instructions	Proper care delivery and reduced complaints
Capacity Management	Digital booking system with room assignments	Better resource utilization

5.5 Design Principles Applied

- **User-Centered Design**
 - Clear information architecture based on user workflows
 - Minimal cognitive load with intuitive navigation
 - Role-based interfaces tailored to each stakeholder
- **Accessibility & Inclusivity**
 - Material 3 design system with WCAG compliance
 - Readable fonts with sufficient contrast ratios
 - Alternative text for images and icons
 - Support for multiple languages (using intl package)
- **Reliability & Trust**
 - Secure Firebase authentication (email/password with history)
 - HTTPS encryption for all data transmission
 - Cloud Firestore with real-time sync for data consistency

- Offline capability for critical data
- Efficiency
 - One-tap bookings with saved preferences
 - Notification-based workflow triggers
 - Searchable provider directories with filters
 - Quick access to recent pets/providers
- Transparency
 - Clear pricing with no hidden fees
 - Real-time booking status updates
 - Complete pet health history visibility
 - Honest review and rating system
- Customization
 - Multiple service types (vet, sitter, hotel)
 - Role-based features and visibility
 - Customizable pet profiles and health tracking
 - Personalized dashboard views

5.6 Key Features Addressing Pain Points

Pet Owner Pain Points:

-  Scattered pet records →  Centralized health database
-  Phone-based scheduling →  Digital booking system
-  Trust issues with sitters →  Verified profiles with reviews and real-time updates
-  Difficult health tracking →  Automated reports and activity logs

Provider Pain Points:

-  Scattered bookings →  Unified calendar system
-  Manual scheduling →  Automated availability management
-  Poor client communication →  In-app messaging and notifications
-  Difficulty building reputation →  Professional profiles with ratings

6.0 Technical Architecture

6.1 Technology Stack

Frontend:

- Flutter 3.x for cross-platform mobile development
- Material 3 design system for modern UI
- flutter_bloc for state management
- Responsive layouts for various device sizes

Backend:

- Firebase Authentication (Email/Password with custom password history)
- Cloud Firestore for real-time database
- Firebase Storage for file management (photos, documents)
- Cloud Functions (Node 18 + TypeScript) for business logic
- Firebase Messaging for push notifications

DevOps & Tools:

- GitHub for version control
- Firebase CLI for deployment
- FlutterFire for configuration management
- Local notifications plugin for reminder system

6.2 Database Schema Overview

Collections:

- `users` - User profiles (pet owners, vets, sitters)
- `pets` - Pet information and medical records
- `bookings` - Appointment and service bookings
- `activities` - Daily logs and activities
- `reports` - Generated health and activity reports
- `vet\_profiles` - Veterinarian specializations and schedules
- `sitter\_profiles` - Sitter services and availability
- `hotel\_stays` - Hotel booking and check-in records

7.0 Implementation Status

7.1 Completed Features

-  Authentication system with role-based login
-  Pet profile CRUD operations with photo upload
-  Vet browsing and booking functionality
-  Pet sitter browsing and booking system
-  Hotel check-in/check-out flow
-  Activity logging with timestamps
-  Health report generation and export (PDF)
-  Push notification infrastructure
-  Local notification reminders
-  Firebase integration with security rules
-  Cloud Functions for business logic
-  Profile editing and photo management

7.2 Production Readiness

Security Implemented:

- Firebase security rules for data access control
- Password history enforcement via Cloud Functions
- Email verification for accounts
- Firebase App Check for API protection

Testing:

- BLoC and widget testing framework set up
- Unit tests for business logic
- Integration tests available via `flutter test`

Deployment Ready:

- Clean architecture with separation of concerns
- Responsive design tested on multiple screen sizes
- Error handling and validation throughout

8.0 Future Enhancements

8.1 Phase 2 Features

- Payment Integration - Stripe or PayPal integration for booking payments
- Advanced Analytics - Pet health trend analysis and predictive recommendations
- AI-Powered Recommendations - Suggest vets/sitters based on pet type and history
- Telemedicine - Video consultation capability for vets
- Multi-language Support - Localization for international markets
- Social Features - Pet community, forums, and user connections

8.2 Technical Improvements

- Offline-First Architecture - Enhanced offline capabilities with local database
- Performance Optimization - Image caching, lazy loading, pagination
- Analytics Integration - Firebase Analytics for user behavior tracking

- Machine Learning - Predict appointment no-shows, recommend services
- Mobile Payment - Integrated payment wallets (Apple Pay, Google Pay)

9.0 Conclusion

PetPal represents a comprehensive solution to a critical gap in pet care management. By unifying pet owners, veterinarians, and service providers on a single platform, the application streamlines the entire pet care lifecycle while building trust through transparent communication and verified professional services.

The design prioritizes user needs across all stakeholder groups, with a clean, intuitive interface that minimizes friction in critical workflows like appointment scheduling and activity tracking. The modern tech stack ensures scalability, security, and real-time updates that are essential for a service-oriented application.

With a solid foundation of implemented features and a clear roadmap for future enhancements, PetPal is positioned to become the go-to platform for comprehensive pet care management.

10.0 Github Repository link

https://github.com/ammaribrahim95/MobileApp_PetPal



Scan for Github Repository